Lesson 03: Experimental and Clinical Psychologists

English Stage 4 (Psychology) · Unit 1: Psychology as a Science

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 1.6 Experimental and Clinical Psychologists; 1.7 The Science of Psychology (Summary).

In this lesson

You will learn who conducts research in psychology, what the clinical practice of psychology is, and why clinicians need science. You will then review all of Chapter 1 through the book's exercises. On the language side, you will practice reporting a debate with reporting verbs and adding emphasis with *not only … but also*.

Learning goals

After this lesson you should be able to:

- Describe who conducts psychological research, where they work and why.
- Define the clinical practice of psychology and explain why research is relevant to it.
- Define an empirically supported treatment and give examples.
- Report two sides of a debate with reporting verbs (*argue, claim, believe, suggest, demonstrate*).
- Apply the ideas of Chapter 1 to new claims and questions.

Key language

Term or phrase	Meaning	Example
doctor of philosophy (PhD)	the highest degree in most academic fields, including psychology; scientific researchers typically have it	Most researchers have a PhD or a master's degree.
subfield	a branch of a discipline	cognitive, developmental, personality, social psychology
experimental psychologist	in this chapter, a psychologist who conducts scientific research, in any subfield	a cognitive psychologist who runs studies at a university
clinical practice of psychology	the diagnosis and treatment of psychological disorders and related problems	the work of clinical and counseling psychologists
empirically supported treatment	a treatment shown by scientific research to result in greater improvement than no treatment, a placebo, or some alternative treatment	exposure therapy for phobias
placebo	an inactive treatment used for comparison, such as a sugar pill	A new therapy should do better than a placebo.

Term or phrase	Meaning	Example
scientifically literate	able to read and evaluate new research and make decisions based on the best available evidence	Clinicians should be scientifically literate.
It is important to consider ...	stresses a point the reader must think about	It is important to consider the relationship between research and practice.
On both sides of the debate, however, there is agreement that ...	reports what opponents agree on	... a scientific approach is essential.
not only ... but also ...	adds a second, stronger point	Not only should research continue, but clinicians should also read it.

Rules and explanations

The ideas

Who conducts research? (1.6)

- People with doctoral degrees (usually the PhD) and master's degrees, helped by research assistants. Some work for government agencies, associations, non-profits or companies, but the majority are college and university faculty (teaching and research staff), who often do research with their students.
- Most are not licensed clinicians: their expertise is in subfields such as behavioral neuroscience and cognitive, developmental, personality or social psychology.
- They enjoy the intellectual and technical challenge and the satisfaction of contributing to knowledge.

Clinical psychology and science

- Psychology also applies research to "help people, organizations, and communities function better" (APA, 2011). Its best-known application is **clinical practice**.
- **Main point:** psychological disorders are part of the natural world, so questions about their nature, causes and consequences are empirically testable.
- Example: popular books claim that adult children of alcoholics have a distinct personality profile, but they are no more likely than anybody else to have these problems (Lilienfeld et al., 2010).
- Whether a therapy works is also empirical: if a new psychotherapy is effective for depression, depressed people who receive it should improve more than a similar group who do not (or who receive another treatment).

Empirically supported treatments. Many forms of psychotherapy (treatment by a therapist rather than with drugs) have strong support and can be as effective as standard drug therapies:

Treatment	Strong support for
Acceptance and commitment therapy (ACT)	depression, mixed anxiety disorders, psychosis, chronic pain, obsessive-compulsive disorder
Behavioral couples therapy	alcohol use disorders
Cognitive behavioral therapy (CBT)	many disorders, including eating disorders, depression and anxiety disorders
Exposure therapy	post-traumatic stress disorder and phobias
Exposure therapy with response prevention	obsessive-compulsive disorder
Family-based treatment	eating disorders

A debate. Many clinical psychologists argue that their field has not paid enough attention to research (for example, by not using empirically supported treatments); others believe these claims are exaggerated (Norcross, Beutler, & Levant, 2005). Both sides agree that a scientific approach is essential; so, the authors conclude, even clinicians who never conduct a study must be **scientifically literate**.

Language focus

1. Reporting a debate. The reporting verb shows how the writer sees a claim.

- *claim* (states; the writer may doubt it): "thousands of websites **claim** that adult children of alcoholics have a distinct personality profile"
- *demonstrate* (shows with evidence): "scientific research **has demonstrated** that ..."
- *argue* (gives reasons for a position) and *suggest* (proposes): "Many ... **have argued** that their field has not paid enough attention to scientific research ... and **have suggested** a variety of changes"
- *believe* (holds an opinion): "Others **believe** that these claims are exaggerated"

A balanced structure: *Many ... argue that ... Others believe that ... On both sides of the debate, however, there is agreement that ...*

2. Emphasis. "So **not only is it important** for scientific research in clinical psychology to continue, **but it is also important** for clinicians ... to be scientifically literate."

> **Note for Arabic speakers** - When *not only* begins a clause, use question word order: "Not only **is it** important ...", "Not only **should research** continue ...", not "Not only it is important".

Practice

1. Reading questions (Section 1.6)

1. Where do the majority of researchers in psychology work, and are most of them licensed clinicians?
2. Why are questions about psychological disorders empirically testable?
3. What do popular books claim about adult children of alcoholics, and what has research shown?

4. What result would show that a new psychotherapy is an effective treatment for depression?
5. Which two empirically supported treatments does the book list for obsessive-compulsive disorder?

2. Book exercises (Section 1.7: the exercises for all of Chapter 1)

1. Practice: Consider three things you know and determine how you acquired that knowledge (authority, intuition, rationalism, empiricism, the scientific method).
2. Practice: Try to generate different research questions to describe, predict, and explain a phenomenon that interests you.
3. Practice: Based on your own experience or on things you have already learned about psychology, list three basic research questions and three applied research questions of interest to you.
4. Practice: List three empirical questions about human behavior. List three nonempirical questions about human behavior.
5. Practice: For each of the following intuitive beliefs about human behavior, list three reasons that it might be true and three reasons that it might not be true: (a) You cannot truly love another person unless you love yourself. (b) People who receive "crisis counseling" immediately after experiencing a traumatic event are better able to cope with that trauma in the long term. (c) Studying is most effective when it is always done in the same location.
6. Discussion: Consider the following psychological claim. "People's choice of spouse is strongly influenced by their perception of their own parents. Some choose a spouse who is similar in some way to one of their parents. Others choose a spouse who is different from one of their parents." Is this claim falsifiable? Why or why not?
7. Discussion: People sometimes suggest that psychology cannot be a science because either (a) human behavior cannot be predicted with perfect accuracy or (b) much of its subject matter (e.g., thoughts and feelings) cannot be observed directly. Do you agree or disagree with each of these ideas? Why? (Discuss it with your tutor first.)
8. Discussion: Some clinicians argue that what they do is an "art form" based on intuition and personal experience and therefore cannot be evaluated scientifically. Write a paragraph about how satisfied you would be with such a clinician and why from each of three perspectives: (a) a potential client of the clinician; (b) a judge who must decide whether to allow the clinician to testify as an expert witness in a child abuse case; (c) an insurance company representative who must decide whether to reimburse the clinician for their services.
9. Practice: Create a short list of questions that a client could ask a clinician to determine whether they pay sufficient attention to scientific research.

3. Language exercise

A. Complete with *believe, claim, demonstrated* or *suggested*.

1. Thousands of websites ____ that adult children of alcoholics have a distinct personality profile.
2. Scientific research has ____ that they are no more likely to have these problems than anybody else.
3. Many clinical psychologists have ____ a variety of changes in the way clinicians are trained.
4. Others ____ that these claims are exaggerated.

B. Join into one sentence beginning with *Not only*: "Scientific research in clinical psychology should continue. Clinicians should also be scientifically literate."

Speak with your tutor

- "Explain what an empirically supported treatment is, using an example from Section 1.6. Correct my definition and my reporting verbs."
- "PSYC 110 Lecture 18 (Mental Illness, Part I) describes a study in which 240 patients with depression received cognitive behavioral therapy, the drug Paxil, or a placebo pill. Ask me why the placebo group matters and what the results suggest about therapy compared with a drug."
- "Role-play: you are a clinician who says therapy is an 'art form'. I am a client and will ask my questions from Exercise 9."
- "PSYC 110 Lecture 1 names five branches of psychology. Ask me to compare them with the subfields in Section 1.6."

Writing task

Project warm-up. Write one paragraph (150 to 200 words) arguing that advice to students about your topic (by default, sleep and memory before exams; or advice related to your own topic) should be based on research, not on intuition or authority. Include a common claim about the topic, reported with *claim*; why intuition or authority is not enough; what evidence would show that the advice works, following the book's logic for therapies (students who follow it compared with a similar group who do not); and one sentence with *not only ... but also*.

Self-editing checklist:

- Do my reporting verbs show my attitude (*claim, argue, demonstrate*)?
- Did I describe the comparison group that the evidence needs?
- Did I avoid saying that research has shown something I have not read?
- Is the word order correct after a fronted *Not only*?
- Does my paragraph have a clear topic sentence and conclusion?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- "Open Access Explained!" (PHD Comics), from the Chapter 1 exercises: https://youtu.be/L5rVH1KGBCY
- Psychological treatments with research support (Society of Clinical Psychology, APA Division 12; the book's div12.org link now leads here): https://societyofclinicalpsychology.org/psychological-treatments/
- Research Digest, the British Psychological Society's blog (moved from the book's address): https://www.bps.org.uk/research-digest
- Note: the book's Scott Lilienfeld video is no longer available, and researchblogging.org now leads to an unrelated website.

Answer key

1. Reading questions

1. Most are college and university faculty (others work for government agencies, associations, non-profits or the private sector). Most are not licensed clinicians; their expertise is in other subfields (1.6, first paragraph).
2. Psychological disorders are part of the natural world, so questions about their nature, causes and consequences can be studied scientifically (1.6).
3. They claim a distinct profile: low self-esteem, feelings of powerlessness and difficulties with intimacy. Research has demonstrated that these adults are no more likely to have these problems than anybody else (1.6).
4. Depressed people who receive it improve more than a similar group who do not receive it, or who receive an alternative treatment (1.6).
5. Acceptance and commitment therapy (ACT) and exposure therapy with response prevention (1.6, list).

2. Book exercises (example answers; many answers are possible)

1. "My friend is upset with me": intuition. "Exercise improves mood": authority (my doctor). "Women and men speak about the same number of words a day": the scientific method (Mehl's study).
2. Procrastination. Describe: How common is it among first-year students? Predict: Do students who procrastinate more get lower grades? Explain: Does fear of failure cause procrastination?
3. Basic: Why do we forget dreams so quickly? Do people remember faces better than names? Are first-born children more talkative? Applied: Does banning phones in class improve exam results? Which therapy best reduces exam anxiety? How can hospitals reduce errors by tired night-shift nurses?
4. Empirical: Do people who exercise report less anxiety? Do children who are read to daily learn to read earlier? Do people in a hurry help strangers less? Nonempirical (values): Is it wrong to lie to protect a friend? Should parents be allowed to spank their children? Is it better to be happy or to be good?
5. (a) True: people who dislike themselves may doubt that anyone loves them, become too dependent, or fail to set healthy limits. Not true: many people with low self-esteem have loving relationships; parents with low self-esteem still love their children; love may increase self-love (the cause could run the other way). (b) True: early support may reduce isolation, teach coping skills and lead to further help. Not true: many people may recover anyway; talking too early may be more upsetting; people may be pushed to talk before they are ready. (c) True: the place may become a memory cue; a familiar place may have fewer distractions; a fixed routine makes it easier to start. Not true: exams are taken elsewhere; varied places may give more cues; one place may become boring. Only systematic research can decide.
6. Not falsifiable as stated: whether people choose a spouse similar to a parent or different from a parent, the result is consistent with the claim, so no observation could count against it (compare

the ESP example in 1.3). It would become falsifiable if it predicted *which* people choose similar spouses.

7. Disagree with both. (a) Predictions "will not be 100% accurate" but can have "greater than chance accuracy" (1.4); a science is defined by its general approach, not by perfect accuracy (1.3). (b) Psychology studies thoughts and feelings through what can be observed systematically, such as the words people speak (Mehl's recordings) or how much patients improve. Some people partly agree with (b), because such observations are indirect.
8. (a) Client: not satisfied; I want a treatment shown to work better than no treatment or a placebo, not one based on intuition, which can be biased. (b) Judge: very dissatisfied; expert testimony in a child abuse case must rest on evidence others can check, and an "art form" cannot be evaluated. (c) Insurance representative: not satisfied; the company should pay for treatments known to be effective, and there is no evidence that these services produce improvement.
9. Is your treatment empirically supported for my problem? What research shows that it works, and compared with what? How do you keep up with new research? How will we measure my progress? What other supported treatments exist for my problem?

3. Language exercise

A. 1. claim. 2. demonstrated. 3. suggested. 4. believe.

B. Not only should scientific research in clinical psychology continue, but clinicians should also be scientifically literate.

My notes